Inequalities

- **Exercise 1**
  Let $f(x)=\dfrac{2x-1}{x+3}$. Find the preimage $f^{-1}\left((1,3)\right)$ of $(1,3)$.
  - **Solution 1.1**
    We have

$$\begin{aligned}
f^{-1}\left((1,3)\right) &= \{x\in\mathbb{R}\setminus\{-3\} : f(x)\in(1,3)\} \\
&= \left\{x\in\mathbb{R}\setminus\{-3\} : 1<\frac{2x-1}{x+3}<3\right\} \\
&= \left\{x\in\mathbb{R}\setminus\{-3\} : 1<\frac{2x-1}{x+3}\right\}\cap\left\{x\in\mathbb{R}\setminus\{-3\} : \frac{2x-1}{x+3}<3\right\} \\
&= \left\{x\in\mathbb{R}\setminus\{-3\} : 0<\frac{2x-1}{x+3}-1\right\}\cap\left\{x\in\mathbb{R}\setminus\{-3\} : \frac{2x-1}{x+3}-3<0\right\} \\
&= \left\{x\in\mathbb{R}\setminus\{-3\} : 0<\frac{x-4}{x+3}\right\}\cap\left\{x\in\mathbb{R}\setminus\{-3\} : \frac{-x-10}{x+3}<0\right\} \\
&= \left((-\infty,-3)\cup(4,\infty)\right)\cap\left((-\infty,-10)\cup(-3,\infty)\right) \\
&= (-\infty,-10)\cup(4,\infty).
\end{aligned}$$

- **Exercise 2**
  Suppose $x,y\in\mathbb{R}$ satisfy

$$0\leqslant|x-y|<\varepsilon$$

  for every $\varepsilon>0$. Prove that $x=y$.
  - **Proof 2.1**
    Suppose $x,y\in\mathbb{R}$. Consider $a=|x-y|$. Then $\forall\varepsilon>0, a\geqslant 0, a<\varepsilon$. By Theorem 1.5, we have that $a=0$ $=|x-y|$, that is, $x=y$.

- **Exercise 3**
  Suppose $a\in\mathbb{R}$ satisfies

$$0\leqslant a<\frac{1}{n}$$

  for every $n\in\mathbb{N}$. Prove that $a=0$.
  - **Proof 3.1**
    Suppose $a\in\mathbb{R}$. Suppose $\forall n\in\mathbb{N}, 0\leqslant a<\dfrac{1}{n}$. Suppose $\varepsilon>0$, we need to prove that $a<\varepsilon$. By the Archimedean property, $\exists N_\varepsilon\in\mathbb{N}$ such that $\dfrac{1}{N_\varepsilon}<\varepsilon$. Thus $0\leqslant a<\dfrac{1}{N_\varepsilon}<\varepsilon$. Therefore $0\leqslant a<\varepsilon$. By Theorem 1.5, we have that $a=0$.

- **Exercise 4**
  Suppose $a\in\mathbb{R}$, suppose that for all $\varepsilon>0$, $a<\varepsilon$. Is it true that $a=0$? Prove it or give a counterexample.
  - **Solution 4.1**
    It's false. Consider $a=-1$, then $a<\varepsilon$ holds for all $\varepsilon>0$, but $a\neq 0$.

- **Exercise 5**
  Consider $f(x)=\dfrac{2x+1}{x-5}$. Find the elements in the domain whose image is at a distance less than 2 from 3.
  - **Solution 5.1**
    We have that

$$\begin{aligned}\{x\in\mathbb{R}\setminus\{5\}:\mathrm{d}(f(x),3)<2\} &= \left\{x\in\mathbb{R}\setminus\{5\}:\left|\frac{2x+1}{x-5}-3\right|<2\right\}\\ &= \left\{x\in\mathbb{R}\setminus\{5\}:\left|\frac{2x+1-3(x-5)}{x-5}\right|<2\right\}\\ &= \left\{x\in\mathbb{R}\setminus\{5\}:\left|\frac{-x+16}{x-5}\right|<2\right\}.\end{aligned}$$

By Proposition 2.3(1), this ``is equivalent to:

$$-2<\frac{-x+16}{x-5}<2.$$

We solve this by splitting it into a system of two inequalities

1. For

$$\begin{aligned}-2<\frac{-x+16}{x-5} &\iff \frac{-x+16+2(x-5)}{x-5}>0\\ &\iff \frac{x+6}{x-5}>0\\ &\implies x\in(-\infty,-6)\cup(5,\infty).\end{aligned}$$

2. For

$$\begin{aligned}\frac{-x+16}{x-5}<2 \iff \frac{-x+16-2(x-5)}{x-5}<0 &\iff \frac{-3x+26}{x-5}<0\\ \iff \frac{3x-26}{x-5}>0&\\ \implies x\in(-\infty,5)\cup\left(\frac{26}{3},\infty\right).&\end{aligned}$$

Taking the intersection of these two solution sets yields

$$((-\infty,-6)\cup(5,\infty))\cap\left((-\infty,5)\cup\left(\frac{26}{3},\infty\right)\right)=(-\infty,-6)\cup\left(\frac{26}{3},\infty\right).$$

Therefore, the required elements in the domain are $x\in(-\infty,-6)\cup\left(\frac{26}{3},\infty\right)$.

- **Exercise 6**
  Solve $|2x-5|-\left|\dfrac{x+1}{x-3}\right|\geqslant 1$.
  - **Solution 6.1**
    Consider a necessary condition $|2x-5|-1\geqslant 0$, which yields $2x-5\leqslant -1\vee 2x-5\geqslant 1 \iff x\leqslant 2\vee x\geqslant 3$. Notice that the expression is undefined at $x=3$, so the domain is $\mathbb{R}\setminus\{3\}$. The zeroes of the absolute value expressions are $x=\frac{5}{2}$, $x=-1$, and $x=3$. Combined with the necessary condition, we consider the following valid intervals for x:
    1. **Case 1**: $x\leqslant -1$.
       In this interval, $2x-5<0\implies |2x-5|=-2x+5$. For the fraction, $x+1\leqslant 0$ and $x-3<0$, so $\dfrac{x+1}{x-3}\geqslant 0\implies\left|\dfrac{x+1}{x-3}\right|=\dfrac{x+1}{x-3}$. The inequality becomes

$$-2x+5-\frac{x+1}{x-3}\geqslant 1\iff -2x+4-\frac{x+1}{x-3}\geqslant 0\iff\frac{(-2x+4)(x-3)-(x+1)}{x-3}\geqslant 0.$$

       Expanding the numerator: $-2x^2+10x-12-x-1=-2x^2+9x-13$. The inequality is

$$\frac{-2x^2+9x-13}{x-3}\geqslant 0\iff\frac{2x^2-9x+13}{x-3}\leqslant 0.$$

       The discriminant of the quadratic equation $2x^2-9x+13$ is $\Delta=(-9)^2-4(2)(13)=81-104=-23<0$, which means $2x^2-9x+13>0$ for all $x\in\mathbb{R}$. Thus, the inequality depends only on $x-3<0\implies x<3$. Intersecting with $x\leqslant -1$ yields $x\in(-\infty,-1]$.

2. **Case 2**: $-1 < x \leqslant 2$.
   In this interval, $|2x-5| = -2x+5$. For the fraction, $x+1>0$ and $x-3<0$, so $\dfrac{x+1}{x-3} < 0$ $\implies \left|\dfrac{x+1}{x-3}\right| = -\dfrac{x+1}{x-3}$. The inequality becomes

   $$-2x+5+\frac{x+1}{x-3} \geqslant 1 \iff -2x+4+\frac{x+1}{x-3} \geqslant 0 \iff \frac{(-2x+4)(x-3)+(x+1)}{x-3} \geqslant 0.$$

   Expanding the numerator $-2x^2+10x-12+x+1 = -2x^2+11x-11$. The inequality is:

   $$\frac{-2x^2+11x-11}{x-3} \geqslant 0 \iff \frac{2x^2-11x+11}{x-3} \leqslant 0.$$

   Since $x \leqslant 2 < 3$, the denominator $x-3$ is negative, so the numerator must be positive, that is, $2x^2-11x+11 \geqslant 0$.
   The roots of $2x^2-11x+11=0$ are $x = \dfrac{11 \pm \sqrt{121-88}}{4} = \dfrac{11 \pm \sqrt{33}}{4}$. So The solution for the numerator is $x \leqslant \dfrac{11-\sqrt{33}}{4} \vee x \geqslant \dfrac{11+\sqrt{33}}{4}$.
   Intersecting this with the case interval $-1 < x \leqslant 2$ gives $x \in \left(-1, \dfrac{11-\sqrt{33}}{4}\right]$.

3. **Case 3**: $x > 3$.
   In this interval, $2x-5>0 \implies |2x-5| = 2x-5$. For the fraction, $x+1>0$ and $x-3>0$, so $\left|\dfrac{x+1}{x-3}\right| = \dfrac{x+1}{x-3}$. The inequality becomes

   $$2x-5-\frac{x+1}{x-3} \geqslant 1 \iff 2x-6-\frac{x+1}{x-3} \geqslant 0 \iff \frac{2(x-3)^2-(x+1)}{x-3} \geqslant 0.$$

   Since $x>3$, the denominator is positive, so the numerator must be positive, that is, $2x^2-12x+18-x-1 = 2x^2-13x+17 \geqslant 0$.
   The roots are $x = \dfrac{13 \pm \sqrt{169-136}}{4} = \dfrac{13 \pm \sqrt{33}}{4}$. So the solution is $x \leqslant \dfrac{13-\sqrt{33}}{4} \vee x \geqslant \dfrac{13+\sqrt{33}}{4}$.
   Intersecting this with the case interval $x>3$ gives $x \in \left[\dfrac{13+\sqrt{33}}{4}, \infty\right)$.

Combining all three cases, the complete solution set is:

$$(-\infty,-1] \cup \left(-1, \frac{11-\sqrt{33}}{4}\right] \cup \left[\frac{13+\sqrt{33}}{4}, \infty\right) = \left(-\infty, \frac{11-\sqrt{33}}{4}\right] \cup \left[\frac{13+\sqrt{33}}{4}, \infty\right).$$

- **Exercise 7**
  Prove that $\forall \varepsilon > 0, \exists \delta > 0$, such that $\forall x, 0 < |x+3| < \delta \implies |2x+6| < \varepsilon$.
  - **Proof 7.1**
    Suppose $\varepsilon > 0$ and let $\delta := \dfrac{\varepsilon}{2}$. Suppose $0 < |x+3| < \delta$. Then we have:

    $$|2x+6| = 2|x+3| < 2\delta = 2\left(\frac{\varepsilon}{2}\right) = \varepsilon.$$

    Therefore, the implication holds.

- **Exercise 8**
  Prove that $\forall \varepsilon > 0, \exists \delta > 0$, such that $\forall x, 0 < |x-2| < \delta \implies \left|x^3-8\right| < \varepsilon$.
  - **Proof 8.1**
    Suppose $\varepsilon > 0$ and let $\delta := \min\left\{1, \dfrac{\varepsilon}{19}\right\}$. Suppose $0 < |x-2| < \delta$.

Since $\delta \leqslant 1$, we have $|x-2|<1$, which implies:

$$-1 < x-2 < 1 \implies 1 < x < 3.$$

We factor the expression $\left|x^3-8\right|$ as $|x-2| \cdot \left|x^2+2x+4\right|$. To bound the term $\left|x^2+2x+4\right|$, we apply the triangle inequality given $1<x<3$:

$$\left|x^2+2x+4\right| \leqslant |x|^2+2|x|+4 < 3^2+2(3)+4 = 19.$$

Since we also have $|x-2|<\delta \leqslant \dfrac{\varepsilon}{19}$, it follows that:

$$\left|x^3-8\right| = |x-2| \cdot \left|x^2+2x+4\right| < \left(\frac{\varepsilon}{19}\right) \cdot 19 = \varepsilon.$$

Therefore, the statement is proved.

- **Exercise 9**
  Prove that $\forall \varepsilon>0, \exists \delta>0$, such that $\forall x, 0<|x-1|<\delta \implies \left|\dfrac{1}{x^2}-1\right|<\varepsilon$.

  - **Proof 9.1**
    Suppose $\varepsilon>0$ and let $\delta := \min\left\{\dfrac{1}{2}, \dfrac{\varepsilon}{10}\right\}$. Suppose $0<|x-1|<\delta$.
    Since $\delta \leqslant \dfrac{1}{2}$, we have $|x-1|<\dfrac{1}{2}$, which implies:

    $$-\frac{1}{2} < x-1 < \frac{1}{2} \implies \frac{1}{2} < x < \frac{3}{2}.$$

    From $\dfrac{1}{2}<x$, we have $x^2>\dfrac{1}{4} \implies \dfrac{1}{x^2}<4$. Also, from $x<\dfrac{3}{2}$, we have $|x+1| \leqslant |x|+1 < \dfrac{3}{2}+1 = \dfrac{5}{2}$.
    Now, we algebraicly simplify and factor the target expression

    $$\left|\frac{1}{x^2}-1\right| = \left|\frac{1-x^2}{x^2}\right| = \frac{|1-x| \cdot |1+x|}{x^2} = |x-1| \cdot \frac{|x+1|}{x^2}.$$

    Using our bounds $\dfrac{1}{x^2}<4$ and $|x+1|<\dfrac{5}{2}$, we obtain that

    $$\frac{|x+1|}{x^2} < 4 \cdot \frac{5}{2} = 10.$$

    Since we also have $|x-1|<\delta \leqslant \dfrac{\varepsilon}{10}$, it follows that:

    $$\left|\frac{1}{x^2}-1\right| = |x-1| \cdot \frac{|x+1|}{x^2} < \left(\frac{\varepsilon}{10}\right) \cdot 10 = \varepsilon.$$

    Therefore, the statement is proved.